11/02/2026

Code-A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 NCERT Booster Program for NEET-2026_NBTS-06A **Time : 180 Min.**

Topics Covered:
Complete Syllabus of NEET-2026

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry, Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A car accelerates from rest at constant rate of 2 m/s^2 for some time after which it decelerates at a constant rate of 3 m/s^2 to come to rest. If total time taken for the motion is 40 seconds then maximum velocity achieved by the car during motion is

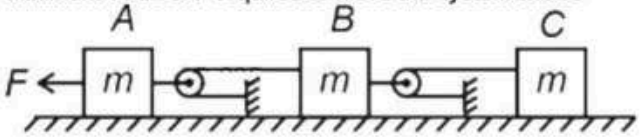
(1) 40 m/s

(2) 48 m/s

(3) 20 m/s

(4) 60 m/s

2. Three blocks *A*, *B* and *C* are placed on a rough horizontal surface as shown in figure. Coefficient of friction between blocks and surface is μ. A force *F* is applied on block *A*. Minimum value of *F* required to move the system will be



(1) 2μ*mg*

(2) 3μ*mg*

(3) 7μ*mg*

(4) 1μ*mg*

3. A ball of mass 2 kg initially moving with a velocity $\left(2\hat{i}+\hat{j}\right)$ m/s collides with a ball of mass 4 kg initially at rest. If after the collision the lighter ball comes to rest, then the velocity of the heavier ball after the collision is

(1) $\left(\hat{i}+\frac{\hat{j}}{2}\right)$ m/s

(2) $(\hat{i}+\hat{j})$ m/s

(3) $(2\hat{i}+\hat{j})$ m/s

(4) $(\hat{i}+2\hat{j})$ m/s

4. A binary star system consisting of two stars of masses m_1 and m_2 separated by distance '*r*', the ratio of their kinetic energies $\frac{K_1}{K_2}$ is

(1) $\frac{m_1}{m_2}$

(2) $\sqrt{\frac{m_1}{m_2}}$

(3) $\sqrt{\frac{m_2}{m_1}}$

(4) $\frac{m_2}{m_1}$

5. A hole is made at the bottom of tank filled with water. If total pressure at bottom is 3 atm (1 atm = 10^5 N/m^2) then velocity of efflux is (Consider area of hole too smaller than cross sectional area of container)

(1) 20 m/s

(2) 40 m/s

(3) 50 m/s

(4) 30 m/s

6. An iron bar of length 10 m is heated from 0°C to 100°C. If the coefficient of linear expansion of iron is 10×10^{-6}/°C, the increase in the length of bar is

(1) 0.5 cm

(2) 1.0 cm

(3) 1.5 cm

(4) 2.0 cm

7. A carnot engine has the same efficiency between 800 K to 500 K and x K to 600 K. The value of x is

(1) 1000 K

(2) 960 K

(3) 846 K

(4) 754 K

8. A : All molecules of a gas sample move with the same speed.
R : Average speed of molecules of a gas sample is zero.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

9. A gas is compressed isothermally to half of its initial volume. The same gas is compressed separately through an adiabatic process until its volume is reduced to half. Then (Assume that the initial state is same in both the cases)

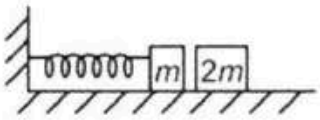
(1) Compressing the gas isothermally or adiabatically will require the same amount of work

(2) Whether compression through isothermal or through adiabatic process requires more work will depend upon the atomicity of the gas

(3) Compressing the gas isothermally will require more work to be done

(4) Compressing the gas through adiabatic process will require more work to be done

10. Two blocks of mass m and $2m$ are kept on smooth horizontal table as shown in figure. Block of mass m is fastend to the spring and $2m$ is kept adjacent to it. If now both the blocks are pushed to the left so that the spring is compressed a distance d. The amplitude of oscillation of block of mass m, long after the system is released, will be



(1) $\frac{d}{\sqrt{3}}$

(2) $d\sqrt{\frac{2}{3}}$

(3) $\frac{2d}{\sqrt{3}}$

(4) $\frac{2d}{\sqrt{2}}$

11. A tuning fork gives 4 beats per second when it vibrates with 60 cm length of a sonometer wire. If it vibrates with 58 cm length of same wire keeping tension in the wire same, it again gives same number of beats per second. Frequency of tuning fork is

(1) 236 Hz

(2) 252 Hz

(3) 258 Hz

(4) 272 Hz

12. **Statement (A):** Speed of sound in air is 330 ms^{-1} at 0°C. If temperature is increased by 1°C then speed of sound in air increases by 0.61 m/s.
Statement (B): Speed of sound depends on amplitude of wave.

(1) Statement A is true but B is false

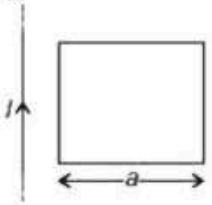
(2) Statement A is false but B is true

(3) Both statements are false

(4) Both statements are true

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13. A square loop of side length *a*, is placed near a long straight wire carrying a current *I* as shown in the figure. Match the phenomenon in column-I with the consequences in column-II.



	Column-I		Column-II
(A)	*I* is increased	(P)	Induced current in the loop is anticlockwise
(B)	*I* is decreased	(Q)	Induced current in the loop is clockwise
(C)	Loop is moved away from the wire	(R)	Wire will attract the loop
(D)	Loop is moved towards the wire	(S)	Wire will repel the loop

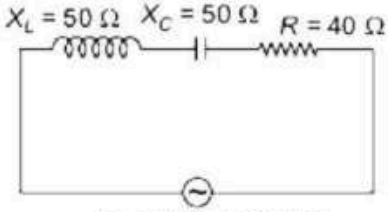
(1) (A)-PR; (B)-QR; (C)-QR; (D)-PS

(2) (A)-PQ; (B)-QR; (C)-QR; (D)-PQ

(3) (A)-PS; (B)-QR; (C)-QR; (D)-PS

(4) (A)-PQ; (B)-QR; (C)-PQ; (D)-PS

14. In the given *LCR* circuit, the effective potential difference across the resistor is (symbols have usual meanings)



$E = (100 \sin 314t)$ V

(1) 100 V

(2) 50 V

(3) $100\sqrt{2}$V

(4) $\frac{100}{\sqrt{2}}$V

15. Which of the following electromagnetic radiations have the largest wavelength?

(1) X-rays

(2) Microwaves

(3) γ-rays

(4) Ultraviolet radiations

16. A photon of energy 3.4 eV is incident on a metal having work function 2 eV. The maximum KE of photo electrons is

(1) 1.4 eV

(2) 1.7 eV

(3) 5.4 eV

(4) 6.8 eV

17. The total energy of electron in the He^+ atom in first excited state is –13.6 eV. The kinetic energy of this electron will be

(1) 13.6 eV

(2) 54.4 eV

(3) 3.4 eV

(4) 27.2 eV

18. The energy equivalent of 1 mg of substance is

(1) 9×10^{10} J

(2) 9×10^{13} J

(3) 9×10^{19} J

(4) 3×10^{13} J

19. In a full wave rectifier, input ac signal has the ripple frequency ν. The output ripple frequency of the current is

(1) $\nu/2$

(2) ν

(3) 2ν

(4) 3ν

20. A screw gauge has main scale divisions of 1 mm each and its circular scale has 50 divisions. While measuring the diameter of a ball, cap end is after 14th marking of main scale and 21st division of circular scale coincides with the reference line. What is the measured diameter of the ball?

(1) 14.40 mm

(2) 14.02 mm

(3) 14.42 mm

(4) 14.44 mm

21. An electric field $\vec{E} = 10x\,\hat{i}$ Vm^{-1} exists in the space, where *x* is in m. Taking potential at (10, 0) m to be zero, the potential at origin will be

(1) 50 V

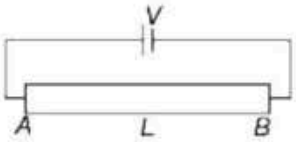
(2) 500 V

(3) 100 V

(4) 1000 V

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22. When an ideal battery of emf V is applied across conductor AB, drift speed of electrons through the conductor is v. If the battery as shown in figure is replaced by an ideal battery of emf $\frac{V}{2}$, the new drift speed of free electrons will be



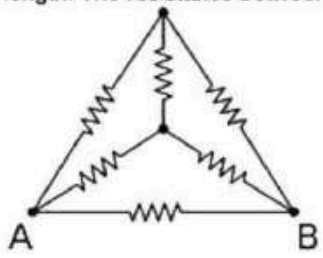
(1) v

(2) $2v$

(3) Zero

(4) $\frac{v}{2}$

23. A uniform wire of resistance 12Ω is bent into a triangular pyramid as shown in figure with each side having same length. The resistance between points A and B is



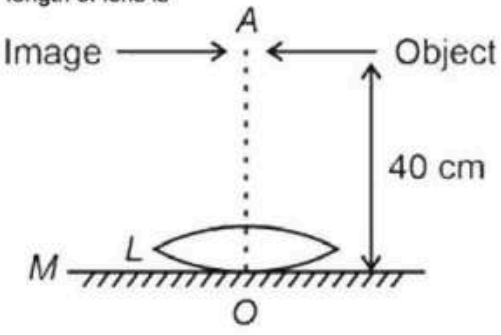
(1) 6Ω

(2) 3Ω

(3) 4Ω

(4) 1Ω

24. A thin convex lens is placed on a plane mirror M. When an object is at A, its real inverted image is formed on A itself as shown in figure. If the distance OA = 40 cm, then the focal length of lens is



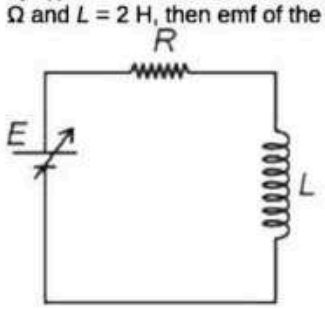
(1) 10 cm

(2) 20 cm

(3) 30 cm

(4) 40 cm

25. In the circuit shown below, the current at any time t is given by $i(t) = 2 + t$, where t is in second and i is in ampere. If $R = 2\ \Omega$ and $L = 2$ H, then emf of the battery at $t = 2$ s is



(1) 6 V

(2) 8 V

(3) 10 V

(4) 12 V

26. If ϵ_0 denotes the permittivity of free space and ϕ_E is the electric flux through the area bounded by the closed surface, Then dimension of $\left(\epsilon_0 \frac{d\phi_E}{dt}\right)$ will be same as of

(1) Electric field

(2) Electric charge

(3) Electric potential

(4) Electric current

27. A body is projected with velocity u from the ground so that horizontal range is $4\sqrt{3}$ times the maximum height attained by it. If horizontal range of projectile is $\frac{\sqrt{n}u^2}{2g}$, then Value of n will be

(1) 2

(2) 3

(3) 4

(4) 5

28. A cylindrical rod of length 1 m and radius 4 cm is mounted vertically. It is subjected to a shear force of 10^5 N at the top. Considering infinitesimally small displacement of top end, the angular displacement of the rod axis from its original position will be. (Shear modulus $= 10^{10}$ N/m^2)

(1) $\frac{1}{4\pi}$ rad

(2) $\frac{1}{40\pi}$ rad

(3) $\frac{1}{160\pi}$ rad

(4) $\frac{1}{400\pi}$ rad

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29. Two different liquids A and B have θ_A and θ_B as angle of contact in a capillary tube. If $K = \frac{\cos\theta_A}{\cos\theta_B}$, then mark correct option.

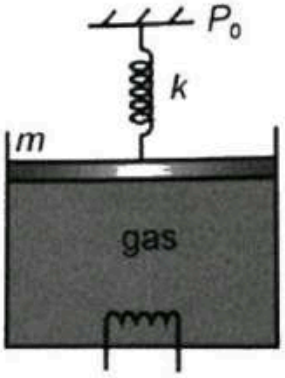
(1) K is negative, then both liquid A and B have convex meniscus

(2) K is negative then both liquid A and B have concave meniscus

(3) K is negative, it implies if A has concave meniscus, then B should have convex meniscus

(4) K is zero, then liquid A has convex meniscus and liquid B has concave meniscus.

30. Heat is supplied by heater so that piston of mass m moves a distance x_0 in upward direction and achieve a speed v_0 then work done by gas is given by (initially spring was in its natural length, A is area of cross section, P_0 is outside pressure)



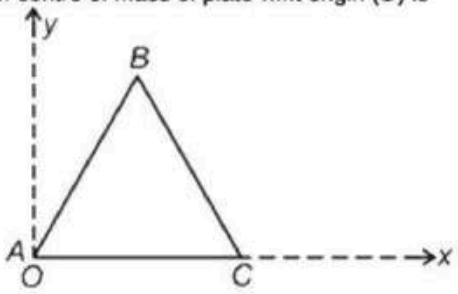
(1) $mgx_0 + \frac{1}{2}mv_0^2$

(2) $\frac{1}{2}mv_0^2 + mgx_0$

(3) $mgx_0 + \frac{1}{2}kx_0^2 + P_0Ax_0 + \frac{1}{2}mv_0^2$

(4) $\frac{1}{2}kx_0 + mgx_0$

31. ABC is an equilateral triangular plate (uniform), having side 'a' lying in $x-y$ plane as shown in figure. Position vector of centre of mass of plate w.r.t origin (O) is



(1) $a\hat{i} + \frac{a}{2}\hat{j}$

(2) $\frac{a}{2}\hat{i} + \frac{a}{2}\hat{j}$

(3) $\frac{a}{2}\hat{i} + \frac{a}{\sqrt{3}}\hat{j}$

(4) $\frac{a}{2}\hat{i} + \frac{a}{2\sqrt{3}}\hat{j}$

32. $\vec{L}$ and $\vec{P}$ represent the angular momentum and linear momentum respectively of a particle of mass 'm'. Position vector of mass 'm' is given as $\vec{r} = a\left(\cos\omega t\hat{i} + \sin\omega t\hat{j}\right)$ here a and ω are dimensional constants. Then direction of force is opposite to

(1) The direction of $\vec{r}$

(2) The direction of $\vec{L}$

(3) The direction of $\vec{P}$

(4) The direction of $\vec{L} \times \vec{P}$

33. A uniform circular disc of radius R has surface mass-density σ. Moment of inertia of this disc about its any diameter will be .

(1) $\frac{\sigma\pi R^2}{4}$

(2) $\frac{\sigma\pi R^4}{4}$

(3) $\frac{\sigma\pi R^4}{2}$

(4) $\frac{\sigma\pi R^2}{2}$

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34. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): The radius vector from Sun to planet sweeps out equal areas in equal time interval and thus areal velocity of planet is constant.
Reason (R): For the central force field angular momentum is conserved.

(1) Both (A) and (R) are correct and (R) is the correct explanation of A

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

35. A block of mass $1\,\text{kg}$, moving along x-axis with speed 10 m/s, enters a rough region ranging from $x = 1$ m to $x = 1.9$ m. The retarding force in this range is $f_r = -10x$ (Newton). Final speed of block as it crosses rough region is (nearly)

(1) $\sqrt{76}$ m/s

(2) $\sqrt{78}$ m/s

(3) $\sqrt{72}$ m/s

(4) $\sqrt{74}$ m/s

36. A small cubical block of mass m is sliding down an inclined plane at an angle 45° with horizontal ground. If acceleration of block is $g/3$, then value of coefficient of kinetic friction between block and inclined plane will be

(1) $\frac{\sqrt{2}}{3}$

(2) $\frac{3-\sqrt{2}}{3}$

(3) $\frac{3+\sqrt{2}}{3}$

(4) $\frac{2-\sqrt{3}}{3}$

37. Two metal spheres of radius R and $2R$ have same surface charge density σ. If they are brought in contact and then separated, the surface charge density on smaller and bigger sphere becomes σ_1 and σ_2. The value of $\frac{\sigma_1}{\sigma_2}$ is

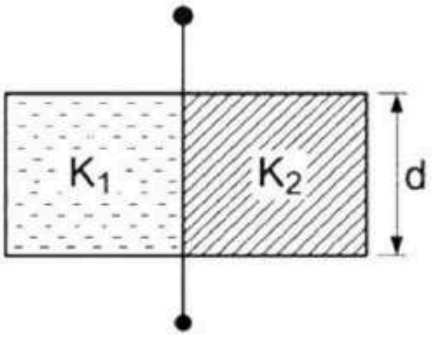
(1) $\frac{1}{4}$

(2) 4

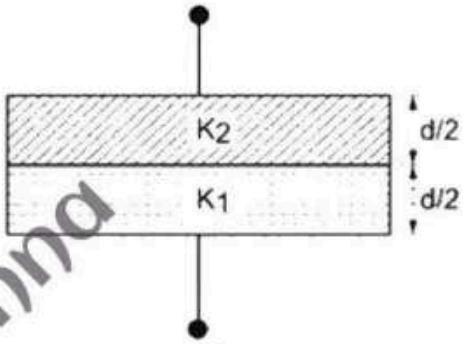
(3) $\frac{1}{2}$

(4) 2

38. A parallel plate capacitor is filled equally (half) with two dielectric constant K_1 and K_2 as shown in figures



(I)



(II)

The distance between plates is d and area of each plate is A. If capacitance in first case and 2nd case are C_1 and C_2 respectively. Then C_1/C_2 is

(1) $\frac{K_1K_2}{K_1+K_2}$

(2) $\frac{\varepsilon_0(K_1+K_2)}{2}$

(3) $\frac{4K_1K_2}{(K_1+K_2)^2}$

(4) $\frac{(K_1+K_2)^2}{4K_1K_2}$

39. There are 10 identical electric bulbs, each is designed to draw a power 20 watt independently from mains supply. They are now Joined in series across main supply. The total power drawn by the combination is

(1) 200 watt

(2) 2 watt

(3) 0.2 watt

(4) 20 watt

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40. Given below are two statements: one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A): Magnetic field lines are continuous and form closed loops
Reason (R): Magnetic monopoles do not exist

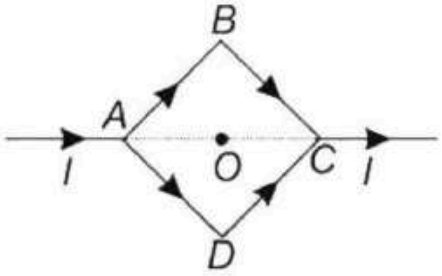
(1) Both $(A)$and (R) are correct but (R) is not correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

41. Figure shows a current carrying square loop *ABCD* of side length 'a' lying in a plane of paper as shown in figure.



If the resistance of path ABC is R and that of ADC path is $2R$, then magnitude of magnetic field (B) at centre of the square loop $ABCD$ is

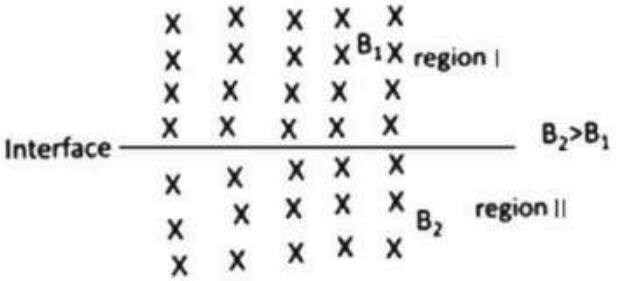
(1) $\frac{2\mu_0 I}{3\pi a}$

(2) $\frac{\sqrt{2}\mu_0 I}{3\pi a}$

(3) $\frac{\mu_0 I}{2\pi a}$

(4) $\frac{3\pi\mu_0 I}{\sqrt{2}a}$

42. Uniform magnetic fields of different strength $B_1$and B_2, both normal to the plane of paper exists as shown in the figure $(B_2 > B_1)$



A charged particle mass $m$and charge q, at the interface at an instant, moves perpendicular into the region (II) with velocity v and returns to interface. It continues to move into region (I) and finally reaches interface. Magnitude of displacement of charge particle during this movement along the interface will be

(1) $\frac{2mv}{qB_1}\left[1-\frac{B_1}{B_2}\right]$

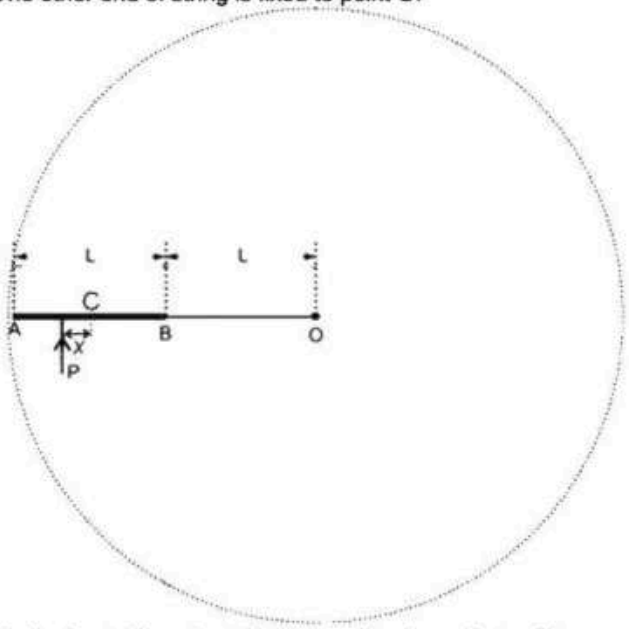
(2) $\frac{mv}{qB_1}\left[1-\frac{B_2}{B_1}\right]$

(3) $\frac{mv}{qB_1}\left[1+\frac{B_1}{B_2}\right]$

(4) $\frac{2mv}{qB_1}\left[1+\frac{B_2}{B_1}\right]$

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43. A thin uniform rod AB of length $L = 18\,\text{cm}$ and mass (m) is kept on a frictionless horizontal table with a massless string of length $L = 18\,\text{cm}$ fixed to one end, as shown in the figure. The other end of string is fixed to point O.



If a horizontal impulse *P* is imparted to the rod at a distance x from the mid point (C) of the rod (as shown in the figure), then the rod AB and string revolve together around the point O with rod remaining aligned with string. Find value of x.

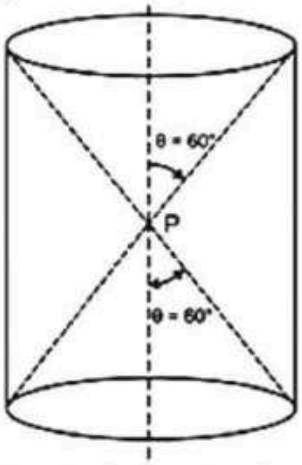
(1) 2 cm

(2) 3 cm

(3) 1 cm

(4) 9 cm

44. A charge Q is kept at the central point P of a closed cylindrical surface, as shown in figure.



If two circular faces form two cones at P as shown in figure whose half angle $\theta = 60^\circ$, the electric flux passes through curved surface of cylinder will be.

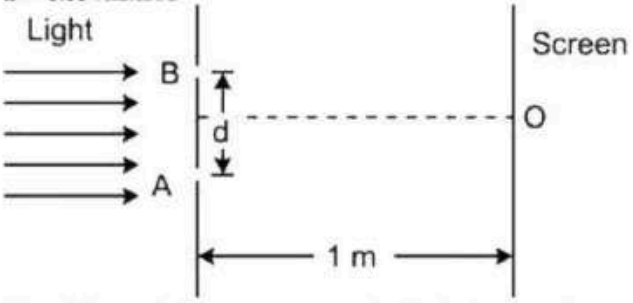
(1) $\frac{Q}{3\varepsilon_0}$

(2) $\frac{Q}{2\varepsilon_0}$

(3) $\frac{2}{3}\frac{Q}{\varepsilon_0}$

(4) Zero

45. In Young's double slit experiment, each of the slit A and B as shown in figure are oscillating about their fixed centre and a with mean separation of 0.8 mm. The distance between the slits at time t is given by $d = (0.8 + 0.04\sin\omega t)$ mm, where $\omega = 0.08$ radian/s



The distance between screen and slits is 1 m and wave length of light used is 6000Å. The interference pattern on the screen changes with time, while central bright fringe (Zeroth order) remain fixed at O. The maximum speed (in μm/s) at which 2nd bright fringe will move is

(1) 2

(2) 6

(3) 4

(4) 3

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CHEMISTRY

46. Correct expression of K_C for the reaction $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ in terms of α is (α is degree of dissociation)

(1) $\frac{\alpha^2}{1-\alpha}$

(2) $\frac{\alpha^2}{4(1-\alpha)}$

(3) $\frac{\alpha^2}{4(1-\alpha)^2}$

(4) $\frac{\alpha}{4(1-\alpha)}$

47. Given below are two statements one is assertion (A) the other is reason (R).
Assertion (A) : Electron gain enthalpy of chlorine is more negative than that of fluorine.
Reason (R) : Due to the small size of fluorine, interelectronic repulsions increase on addition of an electron.
In the light of above statement choose the correct answer.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

48. Mass of $CaCO_3$ required to react completely with 320 mL of 0.25 M HCl is

(1) 4 g

(2) 8 g

(3) 6.5 g

(4) 12.5 g

49. If for the reaction, A(l) → B(g) at 300 K
$\Delta H = 5.7$ kcal mol^{-1}, $\Delta S = 40$ cal K^{-1} mol^{-1}, then ΔG is (R = 2 cal K^{-1} mol^{-1})

(1) –2.5 kcal mol^{-1}

(2) –3.4 kcal mol^{-1}

(3) –6.3 kcal mol^{-1}

(4) –5.1 kcal mol^{-1}

50. Propyne when passed through hot iron tube at 873 K, it produces an aromatic compound X. The compound X is

(1) Benzene

(2) Toluene

(3) Xylene

(4) Mesitylene

51. Which among the following is an actinoid?

(1) Cu

(2) Te

(3) Ce

(4) Th

52. In Carius method of estimation of halogen, 0.5 g of an organic compound gave 0.47 g of AgBr. Percentage of bromine in compound is (molar mass of AgBr = 188)

(1) 20%

(2) 40%

(3) 50%

(4) 80%

53. Due to inert pair effect, the most stable oxidation state of lead is (Atomic number of Pb = 82)

(1) –2

(2) +2

(3) –4

(4) +4

54. Which among the following sets of quantum numbers is not possible?

(1) n = 4, l = 2, m = –2, s = $+\frac{1}{2}$

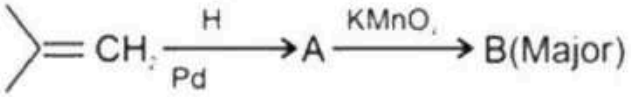
(2) n = 4, l = –2, m = 2, s = $-\frac{1}{2}$

(3) n = 4, l = 2, m = 2, s = $-\frac{1}{2}$

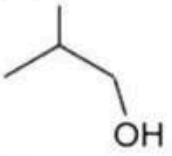
(4) n = 4, l = 2, m = 0, s = $+\frac{1}{2}$

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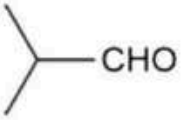
55. Consider the following reaction sequence

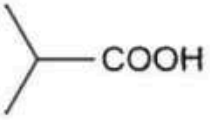


Major product B is

(1) 

(2) 

(3) 

(4) 

56. For the cell reaction

$Ni(s) + 2Ag^{+}(aq) \rightleftharpoons Ni^{2+}(aq) + 2Ag(s)$

$E^{\circ}_{Cell} = 1.05$ V. The standard Gibbs energy $(\Delta_r G^{\circ})$ of the cell reaction is (F is faraday constant)

(1) 1.05 F

(2) 2.1 F

(3) – 1.05 F

(4) – 2.1 F

57. Given below are some aquated transition metal ions

(a) Sc^{3+}

(b) Ti^{4+}

(c) Cr^{3+}

(d) Mn^{2+}

which of them are colourless.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (d) only

(4) (a), (b) and (c) only

58. In a reaction $3A + B_2 \rightarrow A_3B_2$

if 180 atoms of A and 100 molecules of B_2 react, then

(1) B_2 is limiting reactant and 100 molecules of A_3B_2 will be formed

(2) A is limiting reactant and 60 molecules of A_3B_2 will be formed

(3) A is limiting reactant and 180 molecules of A_3B_2 will be formed

(4) B_2 is limiting reactant and 60 molecules of A_3B_2 will be formed

59. Solubility product constant (K_{SP}) of salts of types MX, MX_2 and MX_3 at temperature T are 4.0×10^{-8}, 3.2×10^{-14} and 2.7×10^{-15}, respectively, Solubilities (mole dm^{-3}) of the salts at temperature 'T' are in the order

(1) $MX_3 > MX > MX_2$

(2) $MX_3 > MX_2 > MX$

(3) $MX_2 > MX_3 > MX$

(4) $MX > MX_3 > MX_2$

60. Given below are two statements: One is labelled as **Assertion A** and other is labelled as **Reason R**.

Assertion A : Energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium.

Reason R : Energies of the orbitals in the same subshell increases with increase in the atomic number.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

61. Consider the following statements.

a. BeH_2 and BCl_3 have incomplete octet on central atoms.

b. N_2O is an odd electron molecule.

c. PF_5 and SF_6 have expanded octet on central atom.

The correct statements are

(1) a and b only

(2) b and c only

(3) a and c only

(4) a, b and c

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62. Given below are two statements.
Statement I: SiF_6^{2-} exists but $SiCl_6^{2-}$ does not.
Statement II: Si is more electronegative than C.
In the light of above statements, choose the **correct** option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

63. Number of antibonding electrons in O_2^+ and bonding electrons in B_2 respectively are

(1) 5 and 6

(2) 7 and 4

(3) 5 and 4

(4) 6 and 4

64. Oxidation states of S in $Na_2S_2O_3$, SF_6 and SO_2 are respectively

(1) +3, +6, +2

(2) +2, +6, +3

(3) +2, +6, +4

(4) +3, +6, +4

65. Consider the following statements
(a) Gallium is also known as Eka-silicon
(b) Fourth period of modern periodic table contains 18 elements
(c) The elements after uranium are called transuranium elements
The correct statements are

(1) (b) and (c) only

(2) (a) and (b) only

(3) (a) and (c) only

(4) (a), (b) and (c)

66. The hybridisation and basicity of orthoboric acid are respectively

(1) sp^2, 3

(2) sp^2, 1

(3) sp^3, 1

(4) sp^3, 3

67. Select correct statements among the following
(i) Hyperconjugation involves σ –p conjugation
(ii) Resonance effect operates through sigma bond
(iii) Resonance effect is stronger than inductive effect
(iv) Inductive effect is a temporary effect
Choose the correct option.

(1) (i), (ii), and (iii) only

(2) (i) and (iii) only

(3) (ii) and (iii) only

(4) (i), (ii) and (iv)

68. Among the following, the species which is aromatic, is

(1) 

(2) 

(3) 

(4) 

69. Consider separate solutions of 0.500 M C_2H_5OH (aq), 0.100 M $Mg_3(PO_4)_2$(aq), 0.250 M KBr(aq) and 0.125 M Na_3PO_4(aq) at 25°C. Assume all salts to be strong electrolytes. The correct order of osmotic pressure is

(1) $C_2H_5OH > KBr > Na_3PO_4 > Mg_3(PO_4)_2$

(2) $C_2H_5OH > Mg_3(PO_4)_2 > KBr > Na_3PO_4$

(3) $Na_3PO_4 > KBr > Mg_3(PO_4)_2 > C_2H_5OH$

(4) $C_2H_5OH = Mg_3(PO_4)_2 = KBr = Na_3PO_4$

70. Liquids A and B form ideal solution for all compositions of A and B at 25°C. Two such solutions with 0.25 and 0.5 mole fractions of A in liquid phase have the total vapour pressures of 0.3 and 0.4 bar, respectively. The vapour pressure of pure liquid A is

(1) 0.2 bar

(2) 0.4 bar

(3) 0.6 bar

(4) 0.3 bar

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71. Given below are two statements

Statement I : 0.29g of copper gets deposited at cathode when a solution of $CuSO_4$ is electrolysed for 10 minutes with a 1.5 ampere current.

Statement II : A 38% solution of sulphuric acid is used as an electrolyte in lead storage battery.

In the light of above statements, choose the correct option.

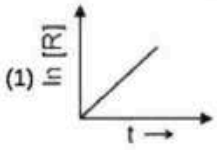
(1) Both statements I and II are correct

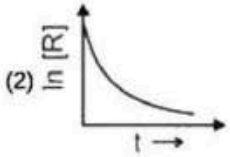
(2) Both statements I and II are incorrect

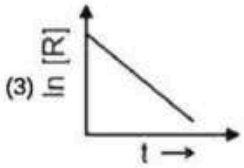
(3) Statement I is correct but statement II is incorrect

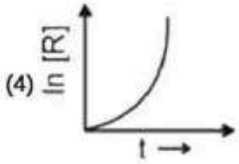
(4) Statement I is incorrect but statement II is correct

72. Identify the correct graph for first order reaction.

(1) 

(2) 

(3) 

(4) 

73. Which among the following is not a mixed oxide?

(1) Fe_3O_4

(2) C_3O_2

(3) Mn_3O_4

(4) Pb_3O_4

74. Which among the following complexes will absorb the maximum wavelength of light (nm)

(1) $[Co(NH_3)_5(H_2O)]^{3+}$

(2) $[CoCl(NH_3)_5]^{2+}$

(3) $[Co(NH_3)_6]^{3+}$

(4) $[Co(CN)_6]^{3-}$

75. Consider the following reaction sequence.

$$CH_3-CH_2-C\equiv CH \xrightarrow[(ii)\ CH_3CH_2Br]{(i)\ NaNH_2} \underset{(Major)}{X}$$

$$\underset{(Major)}{X} \xrightarrow[BaSO_4]{H_2/Pd} \underset{(Major)}{Y} \xrightarrow{\text{Baeyer's reagent}} \underset{(Major)}{Z}$$

X, Y and Z respectively are.

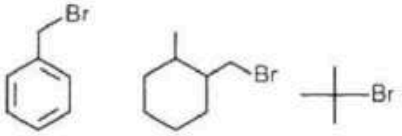
(1) Hexyne, cis-Hex-2-ene and Propanal

(2) Hex-2-yne, trans-Hex-3-ene and Propanoic acid

(3) Hex-3-yne, trans-Hex-2-ene and Propanol

(4) Hex-3-yne, cis-Hex-3-ene and Hexan-3, 4-diol

76. For the following compounds, the correct statement with respect to nucleophilic substitution reaction with HI are



(I) (II) (III)

(A) II follows S_N2 mechanism.

(B) III follows S_N1 mechanism.

(C) I undergoes racemisation in polar protic solvent.

Choose the correct option.

(1) A and B only

(2) A, B and C

(3) A and C only

(4) B and C only

77. Match List-I with List-II

	List-I (Molecule)		**List-II (Boiling point)**
a.	n-Butane	(i)	281 K
b.	Methoxyethane	(ii)	322 K
c.	Propanal	(iii)	370 K
d.	Propan-1-ol	(iv)	273 K

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

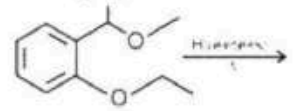
(2) a(iv), b(iii), c(ii), d(i)

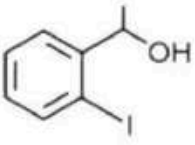
(3) a(iv), b(i), c(ii), d(iii)

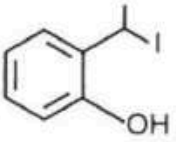
(4) a(ii), b(iv), c(i), d(iii)

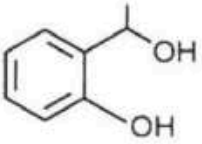
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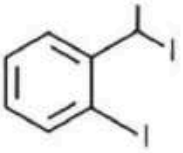
78. The major product formed in the following reaction is



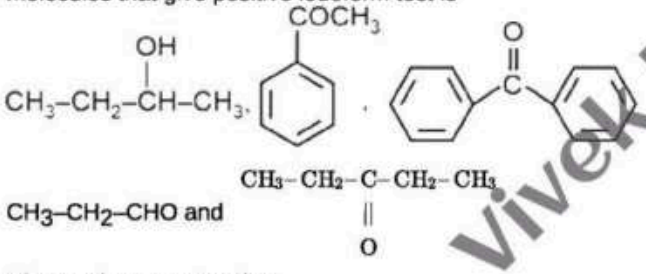
(1) 

(2) 

(3) 

(4) 

79. Consider the following given compounds. The number of molecules that give positive iodoform test is



Choose the correct option.

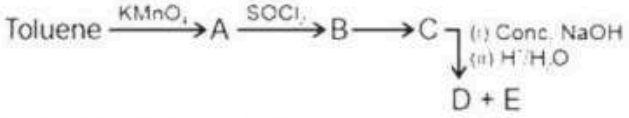
(1) 3

(2) 1

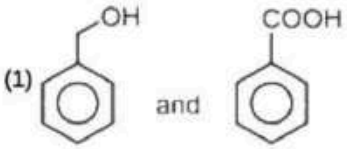
(3) 2

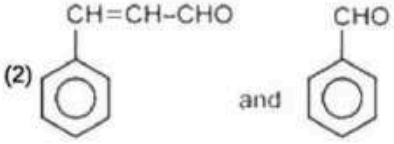
(4) 4

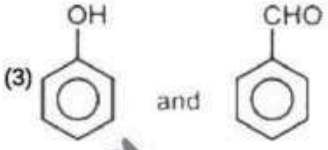
80. In the following sequence of reactions:

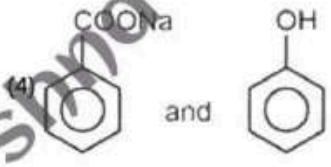


Product E and F respectively are

(1) 

(2) 

(3) 

(4) 

81. Given below are two statements. One is labelled as **assertion (A)** and other is labelled as **reason (R)**.

Assertion (A): In strongly acidic medium, aniline becomes more reactive towards electrophilic reagent.

Reason (R): $-NH_2$ of aniline gets protonated in acidic medium and lone pair of electron is no longer available for resonance.

In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

82. Given below are two statements

Statement I : Glucose on oxidation with nitric acid yields gluconic acid

Statement II : Glucose does not form the hydrogensulphite addition product with $NaHSO_3$

In the light of above statements, choose the correct option.

(1) Both statements I and II are correct

(2) Both statements I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

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83. Identify the colour of the dye formed upon addition of β-naphthol to benzene diazonium salt.

(1) Prussian blue

(2) Yellow

(3) Scarlet red

(4) White

84. Among the hydrides of group 16 elements, the one with highest boiling point is

(1) H_2S because of hydrogen bonding

(2) H_2Se because of high molar mass

(3) H_2O because of hydrogen bonding

(4) H_2Te because of molar mass

85. Match List-I with List-II

	List-I (Oxoacid of P)		**List-II (Basicity)**
a.	H_3PO_4	(i)	Dibasic
b.	H_3PO_3	(ii)	Monobasic
c.	H_3PO_2	(iii)	Tetrabasic
d.	$H_4P_2O_7$	(iv)	Tribasic

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(ii), b(iv), c(iii), d(i)

86. Consider the following cell reaction:

$2Fe(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq.) + 2H_2O(l)$

$E^o = 1.67$ V.

The cell potential at 25°C is

(Given: $[Fe^{2+}] = 10^{-3}$ M, $p_{O_2} = 0.1$ atm and pH = 3)

(1) 1.92 V

(2) 1.63 V

(3) 1.57 V

(4) 1.17 V

87. Two reactions R_1 and R_2 have identical pre-exponential factors.

Activation energy of R_1 and R_2 differs by 20 kJ/mol. The value of $\ln(k_2/k_1)$ at 27°C is

[k_1 and k_2 are rate constants of R_1 and R_2 respectively]

(1) 4

(2) 12

(3) 8

(4) 6

88. Match List-I with List-II

	List-I (Metal ions)		**List-II (Magnetic moment in BM)**
a.	V^{2+}	(i)	5.9
b.	Cr^{2+}	(ii)	3.87
c.	Mn^{2+}	(iii)	4.9
d.	Sc^{2+}	(iv)	1.73

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(ii), b(iii), c(i), d(iv)

89. Which of the following complex species is not expected to exhibit optical isomerism?

(1) $[Co(en)_3]^{3+}$

(2) $cis[Co(en)_2Cl_2]^+$

(3) $[Co(NH_3)_3Cl_3]$

(4) $cis[Co(en)(NH_3)_2Cl_2]^+$

90. A salt is treated with conc. H_2SO_4, brown fumes were evolved which gets dense on heating with Cu turnings. The anion present in salt could be

(1) I^-

(2) Cl^-

(3) NO_3^-

(4) S^{2-}

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BIOLOGY

91. According to R.H. Whittaker's classification system, select the feature which is not similar between the organisms belonging to the kingdoms with heterotrophic mode of nutrition only and organism of the kingdom with cellular body organisation having nuclear membrane.

(1) They absorb soluble organic matter from dead substrates.

(2) They occur in soil as well as in water.

(3) Cell wall of its members are composed of cellulose, galactans, mannans and calcium carbonate.

(4) They show the presence of semi autonomous cell organelle.

92. The layer of cells lies just below the endodermis in dicot root
(a) Forms lateral roots
(b) Cannot be parenchymatous
(c) Is also found in maize stem
(d) Gives rise to a part of vascular cambium ring
(e) May act as storage organ
Choose the **correct** answer from the following options.

(1) (b), (c) and (d) are true

(2) (a), (c) and (e) are true

(3) (a), (b) and (e) are false

(4) (b) and (c) are false

93. How many of the following plants have archegonia?
Dryopteris, Volvox, Marchantia, Cycas, Cedrus, Laminaria, Chlamydomonas, Adiantum, Funaria, Salvinia

(1) Six

(2) Five

(3) Seven

(4) Four

94. Identify the feature which is **not** true for the plants better adapted for dry tropical regions and have Hatch and Slack pathway.

(1) They have large number of chloroplast in cells surrounding vascular bundle in monocot leaf.

(2) They show fixation of carbon dioxide twice.

(3) Primary CO_2 acceptor is a four-carbon compound

(4) They lack the process where RuBisCO shows greater affinity towards oxygen, than CO_2

95. Oxidation of a molecule of glyceraldehyde-3-phosphate in a living cell through glycolysis leads to synthesis of

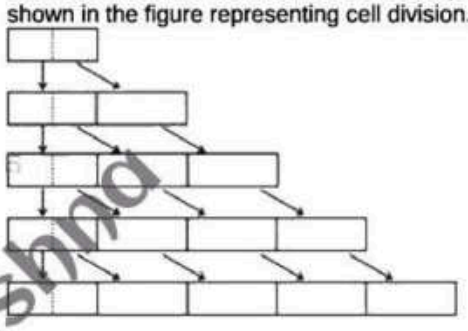
(1) Pyruvic acid without any net gain of ATP molecules

(2) One molecule of ATP, 2 molecules of NADH + H^+, and one molecule of pyruvic acid.

(3) One molecule of pyruvic acid, 2 molecules of ATP and one molecule of NADH + H^+

(4) Pyruvic acid without any net gain of NADH + H^+ molecules.

96. Select the **correct** statement w.r.t. the pattern of growth shown in the figure representing cell division.



(1) This pattern of growth is not shown by shoot elongating at constant rate.

(2) If we plot this type of growth against time, we will get a straight line.

(3) This growth pattern is called geometric growth.

(4) This type of growth occurs during initial development of embryo from zygote.

97. Anatomy of leaf in a member of grass family is similar to that of dorsiventral leaves in many ways but former shows some characteristic difference with the latter ones. These differences are
(a) Mesophyll cells are not differentiated into palisade and spongy parenchyma.
(b) Presence of large, empty and colourless cell on adaxial surface.
(c) Having cuticularized epidermis.
(d) Presence of less stomata on lower epidermis than the upper epidermis.
Choose the **correct** answer from the options given below.

(1) (b) and (c)

(2) (a) and (b)

(3) (c) and (d)

(4) (a) and (d)

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98. Select the **incorrect** statement regarding chromosomal theory of inheritance.

(1) Chromosomes as well as genes occur in pairs in the diploid cell.

(2) Alleles of a gene along with homologous chromosome segregate at the time of gamete formation.

(3) A gamete has only one chromosome and only one allele.

(4) A paired condition of homologous chromosomes as well as Mendelian factors is restored during fertilisation.

99. A phytohormone which does not allow nearby lateral or axillary buds to grow

(1) Also inhibits abscission of young leaves

(2) Was first isolated from fungus *Gibberella fujikuroi*

(3) Stimulates shoot formation on stem cutting

(4) Is the only gaseous phytohormone.

100. 5-methyl uracil is

(1) A purine base

(2) A nine membered double ring structure

(3) Present in mRNA

(4) Present in genetic material of *E. coli*

101. Which among the following statements is **incorrect**?

(1) Methanogens are commonly found in anaerobic sludge during sewage treatment

(2) Biofertilizers are inorganic compound that enrich the nutrient quality of soil.

(3) BOD is an indirect measure of the organic matter present in water.

(4) Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environment which fix atmospheric nitrogen.

102. In *Citrus*, the seeds

(1) Can never be apomictic

(2) Contain haploid embryos only

(3) Have genetically identical embryo except zygotic one

(4) Have coleorhiza as the covering of radicle

103. Match column I with column II and select the **correct** option.

	Column I		**Column II**
A	*Funaria*	(i)	Sexual reproduction takes place through fusion of two gametes that are similar in size and both are non-motile
B	*Spirogyra*	(ii)	The pollen tube carries the male gametes and grows towards archegonia in the ovules
C	*Eucalyptus*	(iii)	Plants of this group have leaf like structure with haploid set of chromosomes in main plant body
D	*Ginkgo biloba*	(iv)	Female gametophyte remain enclosed within ovule that are enclosed by ovary wall

(1) A(iii), B(i), C(ii), D(iv)

(2) A(iii), B(ii), C(iv), D(i)

(3) A(ii), B(i), C(iv), D(ii)

(4) A(iv), B(iii), C(i), D(ii)

104. In an organism following stretch of a polypeptide is synthesized.
Histidine – Cysteine – Methionine – Tryptophan – Aspartic acid
Codons for the given amino acids are as follows:
His – CAU or CAC
Cys – UGU or UGC
Met – AUG
Trp – UGG
Asp – GAU or GAC
Here in the options only one strand of the double stranded DNA has been represented. Then which among the following can be sequence of the template in DNA that codes for the given peptide?

(1) 3' GTCCCACATGCAGTG 5'

(2) 5' GTCCCATACGCAGTG 3'

(3) 5' GTCCCACATGCAGTG 3'

(4) 3' GTCCATCCGGCAGTG 5'

105. The pollen grains are the carrier of male gametes in
(a) The members of group to which *Cedrus* belong
(b) All vascular embryophytes
(c) Plants that have vessels in vascular tissue
(d) All the plants with xylem and phloem in vascular tissue
The **correct** one(s) is/are

(1) (a) and (c)

(2) (b) and (d)

(3) (c) only

(4) (a), (b) and (c)

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106. The type of epithelium that lines the stomach of humans is 'X' and the type of epithelium that lines the intestine and aids in nutrient absorption from the food is 'Y'.
'X' and 'Y' are similar to each other due to the presence of

(1) A single layer of cube-like cells with central nuclei

(2) A single layer of tall and slender cells with basal nuclei

(3) Microvilli on the free surface of cells giving brush-border appearance.

(4) Cilia on the free surface of cells

107. In *Periplaneta americana*, the structures that help in removal of excretory products from haemolymph and are present at the junction of midgut and hindgut are characterised by the
a. Presence of inner epithelium of large flat cells with basal nuclei and an outer sheath of connective tissue
b. Presence of glandular and ciliated cells
c. Conversion of nitrogenous wastes into uric acid
Select the **correct** option.

(1) Only a

(2) Only a and c

(3) Only b and c

(4) a, b and c

108. How many of the structures mentioned in the box given below is/are paired in an adult frog?

Cerebral hemisphere, Cerebellum, Optic lobe, Diencephalon, Medulla oblongata

Choose the **correct** option.

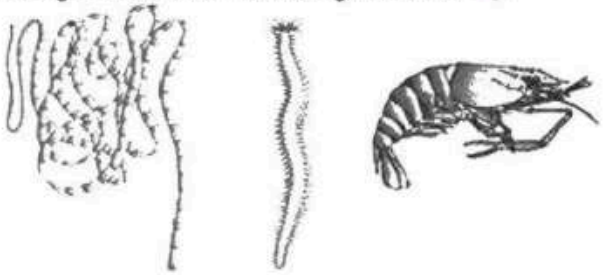
(1) 3

(2) 1

(3) 2

(4) 5

109. The figures of three animals are given below.



'A' 'B' 'C'

Choose the correct option w.r.t them.

(1) Both 'A' and 'B' exhibit metameric segmentation that makes every segment indispensable for survival.

(2) Both 'B' and 'C' possess the same type of circulatory system as present in *Loligo* and *Sepia*.

(3) Both 'A' and 'C' possess the body cavity which is completely lined by mesoderm.

(4) Both 'B' and 'C' are dioecious and free-living.

110. Presence of which of the following features is common between the members of the phyla Ctenophora and Echinodermata?

(1) Bilateral symmetry in adults'

(2) Radial symmetry in larval stage

(3) Indirect development

(4) Presence of mesoderm

111. The enzyme named catalase causes the breakdown of hydrogen peroxide into water and oxygen. From 10°C, its activity doubles for every 10°C rise till 40°C, after which it falls rapidly. Which of the following types of curve is obtained when enzyme activity is plotted against temperature?

(1) Bell-shaped

(2) Linear rise

(3) Rectangular

(4) Sigmoid

112. **Assertion (A)** : The most common tertiary structure of proteins is formed by the union of one α-helix and one β-pleated chain.
Reason (R) : Both α-helix and β-pleated chains of proteins lack hydrogen bonds.
In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

113. Read the given statements and select the **correct** option w.r.t. humans.
Statement (A) : If the alveolar pO_2 is 104 mm Hg and venous blood pO_2 is 40 mm Hg, equilibrium is reached within seconds due to active transport of O_2 from alveoli to venous blood.
Statement (B) : The amount of CO_2 that can diffuse through the diffusion membrane per unit difference in the partial pressure is much higher compared to that of O_2.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is incorrect

(4) Only statement (B) is incorrect

114. If a person's heart rate is 100 beats/min, stroke volume is 50 mL, then calculate the time in which his left ventricle pumps 5 litres of blood into the aorta.

(1) 50 sec

(2) 60 sec

(3) 100 sec

(4) 500 sec

Vivek Krishna

115. Observe the given hypothetical data for a normal human's kidneys

ADH (pg/mL) in blood plasma	Urine volume (mL/day)
0	3000
20	2000
50	800
100	400

On the basis of the above given data, which of the following statements is correct?

(1) The urine concentrating ability depends upon the levels of ADH in blood plasma

(2) A high level of ADH increases the permeability of the distal tubules for water and thus, increases the amount of urine.

(3) When high levels of ADH are present, the fluid at the end of the collecting duct has eventually the same osmolarity as the DCT *i.e.*, about 200 mOsmol/mL.

(4) Large amounts of urea is actively reabsorbed from the inner medullary collecting ducts into the interstitium, when ADH level is low.

116. Read the following Assertion (A) and Reason (R) and select the correct option.

Assertion (A) : In the experiment conducted by T.H. Morgan on *Drosophila*, the genes for body colour and eye colour showed 1.3% recombination, whereas genes for eye colour and wing size showed 37.2% recombination.

Reason (R) : In *Drosophila*, genes for body colour and eye colour are on the same chromosomes whereas gene for wing size is on a different chromosome.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

117. For the development of a typical embryo sac from one megaspore mother cell, what should be the number of generations of meiotic and mitotic divisions respectively?

(1) 1 and 7

(2) 3 and 6

(3) 7 and 1

(4) 1 and 3

118. How many among the following characteristics are common between the prothallus of pteridophyte and thallus of bryophytes?

(a) Gametophytic stage
(b) They are photosynthetic structures
(c) They bear sex organs
(d) Diploid non vascular structures
(e) Free living structures
(f) Presence of archegonia
(g) Spore producing structures

The **correct** ones are

(1) (c), (d) and (g)

(2) (a), (b), (c), (e) and (f)

(3) (b), (d), (e) and (f) only

(4) (a), (c), (d) and (e)

119. Read the following statements w.r.t. the floral formula and diagram and choose the **correct** option.

(1) Fusion of different members of a whorl is indicated by a line drawn above the symbol of the floral parts and adhesion by enclosing the figure within the bracket.

(2) Floral formula represents cohesion and adhesion within the parts of whorls and between the whorls, respectively

(3) A floral diagram provides information about the number of parts of flower and their arrangement but not the relation they have with one another.

(4) ⊕ stands for the zygomorphic nature of the flower and % for actinomorphic nature of flower.

120. In a model plant, if the genes P, Q and R control the flower colour, height of the plant and position of the flowers, respectively. Consider the following genotypes and its corresponding phenotypes:

Flower colour (Gene P)
PP –Red
Pp – Pink
pp – white
Plant height (gene Q)
QQ – Tall
Qq–Intermediate height
qq – Dwarf
Flower Position (R)
RR – Axillary
Rr – Axillary
rr – Terminal

What proportion of plant would be pink axillary flowered intermediate height plants in the progeny if two triple heterozygous plants are crossed?

(1) 6.25%

(2) 12.5%

(3) 18.75%

(4) 33.33%

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121. Read the following Assertion (A) and Reason (R) and select the correct option.
Assertion (A): Failure of segregation of the chromatids of sex chromosome can be observed in human females.
Reason (R): In humans, females have homologous sex chromosome.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

122. Arrange the following steps involved for the synthesis of a polypeptide in the correct order and choose the option accordingly.
A. The anticodon of tRNA recognizes the mRNA codon.
B. hnRNA is made from the DNA by the help of enzyme
C. mRNA binds to the ribosomes.
D. Unwinding of DNA double helix.
E. mRNA leaves the nucleus of the cell.
F. The amino acids of two adjacent tRNA form a peptide bond.

(1) B, D, E, C, A, F

(2) D, E, B, F, C, A

(3) D, E, B, A, C, F

(4) D, B, E, C, A, F

123. Read the following statements and select the correct option.
Statement I : In some cases, per cent cover or biomass is a more meaningful measure of the population size.
Statement II: Predator in nature are not prudent because if they are too efficient and over exploit its prey, then prey might become extinct and following it, the predator will also become extinct for lack of food.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

124. Select the **correct** arrangement of the plant in the ascending order regarding the number of leaves arising at each node.

(1) Guava < *Alstonia* < China rose

(2) Guava < Mustard < China rose

(3) Sun flower < *Calotropis* < *Alstonia*

(4) *Calotropis* < Guava < China rose

125. Read the following statements and select the **correct** option
A. Number of mammals species in the world is more than the combined total of the species of fungi, amphibians, reptiles and fishes.
B. Aquatic ecosystem occupy larger area as compared to terrestrial ecosystems on the Earth.
C. The communities with more species, generally tend to be more stable than those with less species.
D. Large number of species in an ecosystem have higher number of niches, and they will be having more interactions and more interrelationships.

(1) Both statements A and C are true and statement C is the correct explanation of statement A.

(2) Statements C and D are true and statement D explains statement C correctly.

(3) Both statements A and B are true but statement B does not explain statement A.

(4) All A, B, C and D statements are true but none of them are related to each other.

126. Which among the following statements is/are **true** for the lipid component of the plasma membrane?
(a) In the membrane of human erythrocyte, fraction of lipid is more as compared to proteins.
(b) It gives quasi fluid nature to the plasma membrane.
(c) Its head part is hydrophobic.
(d) The tail part is made up of saturated hydrocarbons.
The **correct** one(s) is/are

(1) Only (a)

(2) Only (b) and (d)

(3) (b), (c) and (d)

(4) (a) and (d)

127. Arrange the events occur in the life cycle of the gymnosperm in appropriate order.
A. Male gametes are carried through pollen tube towards archegonia in the ovules.
B. Fusion of male and female gamete lead to the formation of zygote.
C. Development of embryo inside the multicellular female gametophyte within megasporangium.
D. Pollen tube discharge content in the mouth of archegonia.
E. Release of male gametophyte from microsporangium.
Choose the **correct** answer from the options given below.

(1) C, A, E, B, D

(2) E. A, D, B, C

(3) C, E, A, B, D

(4) E, A, C, D , B

128. A diploid eukaryotic cell has three pairs of chromosomes. In which stage the number of chromosomes (represented by n) and DNA content (represented by C) will be 2n and 4C?

(1) Late G_1

(2) Prophase

(3) At the end of anaphase

(4) In the beginning of S phase

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129. An ecosystem has five trophic levels with 10% energy transfer efficiency.
Producers → Primary consumers → Secondary consumers → Tertiary consumers → Top carnivores
If the top carnivore receives 10 kJ energy, what is the minimum NPP required at the producer level?

(1) 1,000 kJ

(2) 10,000 kJ

(3) 100,000 kJ

(4) 1,000,000 kJ

130. A living cell shows structurally normal mitochondrial intact DNA, and functional ribosomes. However, this cell does not produce ATP from glucose by any of the pathways.
Which among the following conditions is responsible for the above scenario?

(1) Oxygen is absent

(2) Cytosolic enzymes are non-functional

(3) Complex II of the inner mitochondrial membrane is less efficient

(4) ATP synthase is transporting protons from inter-membrane space of mitochondria to the matrix

131. Given below are three statements (a), (b) and (c)
(a) It is an auto-immune disorder.
(b) It is characterised by weakening and paralysis of skeletal muscles.
(c) In this disorder, neuro-muscular junction is affected leading to fatigue.
All of the above given statements are true for

(1) Muscular dystrophy

(2) Osteoporosis

(3) Arthritis

(4) Myasthenia gravis

132. The part of human brain stem which is directly connected to spinal cord and also forms the part of hind brain assists in

(1) Integration of visual, tactile and auditory inputs

(2) Conversion of short-term memory into long-term memory

(3) Regulation of gastric secretions and respiration

(4) Controlling urge for eating and drinking

133. **Assertion (A) :** Cortisol and glucagon are antagonistic to each other w.r.t their involvement in maintaining blood glucose levels.
Reason (R) : Cortisol is involved in glycogenesis while glucagon stimulates the process of glycogenolysis.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

134. In a human female, the following changes were recorded during 'X' phase of the menstrual cycle:
• Estrogen levels rise gradually but sharply just before ovulation.
• LH levels surge shortly after estrogen level reaches at its peak.
• Progesterone level remains almost unchanged.
Identify 'X' and choose the correct option.

(1) Menstrual

(2) Follicular

(3) Luteal

(4) Secretory

135. Both 'Saheli' and implant are known to prevent conception. If we compare both of them, then

(1) Latter one contains steroidal preparation while the former one is non-steroidal.

(2) Former one alters the quality of the cervical mucus while the latter one does not.

(3) Latter one increases the phagocytosis of sperm while former one does not

(4) Former one inhibits implantation but latter one does not.

136. A plasmid named pBR322 possesses amp^R and tet^R genes along with the restriction sites for certain restriction enzymes. If a foreign gene of interest is inserted at a site 'X' and then the recombinant plasmid is transformed into *E. coli* and plated on a suitable media, then the following results are obtained

Media	**Growth of recombinants**
Ampicillin	+
Tetracycline	–

The above results indicate that the foreign gene of interest is most likely ligated with the restriction enzyme named

(1) *Cla*I

(2) *Sal*I

(3) *Pvu*I

(4) *EcoR*I

137. Choose the **incorrect** match among the following.

(1)	Pomato	Result of micropropagation
(2)	Somaclones	Genetically identical to the original parent plant
(3)	Explants	Possess the characteristic named totipotency
(4)	RNAi	Method of cellular defence in eukaryotes

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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138. Choose the incorrect statement w.r.t. evolution.

(1) Analogous structures are a result of convergent evolution.

(2) There must be a gentic basis for selection and to evolve.

(3) The rate of appearance of new forms is not linked with the life cycle or the life span.

(4) Fitness is the end result of the ability to adapt and get selected by nature.

139. Comprehend the given statements and select the option that correctly states them as true (T) or false (F).
(a) Morphine, a cannabinoid, is a very effective sedative and painkiller.
(b) Amphetamine and atropine are used as medicines to help patients in treating insomnia.
(c) Smoking increases CO content in the blood.
(d) Excessive dosage of crack causes hallucinations.

	(a)	(b)	(c)	(d)
(1)	T	T	T	T
(2)	T	T	F	T
(3)	T	F	T	F
(4)	F	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

140. After 2nd trimester of pregnancy, the

(1) Corpus luteum becomes hyperactive and secretes hCG more

(2) Pituitary gland secretes LH and FSH due to feedback inhibition of GnRH

(3) Placenta secretes sufficient amounts of progesterone and estrogen

(4) Foetal gonads start secreting hCG

141. If the sequence 5' GAATTC 3' is present in the circular plasmid 'P' for three times and the sequence 5'GGATCC3' is present only once then, how many fragment(s) would be obtained upon complete digestion by *Eco*RI?

(1) Two

(2) Three

(3) One

(4) Four

142. How many of the components given in the box below are present in both areolar as well as adipose tissues?

Fibroblasts, Adipocytes, Macrophages, Mast cells, Chondrocytes

Select the correct option.

(1) Four

(2) Three

(3) Two

(4) Five

143. Choose the **correct** statement w.r.t *Periplaneta americana*.

(1) Presence of a boat-shaped sternum on the 9th abdominal segment in males is used to distinguish them from female members.

(2) Head is triangular and is formed by the fusion of six embryonic segments.

(3) After the last moult of a cockroach nymph, the mandibles become harder and wing pads remain persistent throughout the adult stage.

(4) Its fertilised eggs are encased in ootheca, which is 8 cm long and contains 9–10 eggs.

144. Consider the given features:
(a) Absence of excretory system
(b) Possesses calcareous ossicles
(c) Difference in the body symmetry depending on the stages in their life cycle
(d) Dioecious and exhibits sexual reproduction
The above given features are true for all of the given organisms, **except**

(1) Sea cucumber

(2) Sea lily

(3) Sea hare

(4) Sea urchin

145. *Ichthyophis* and *Bangarus* are similar to *Naja* and *Vipera* in having/exhibiting

(1) Moist skin

(2) Cutaneous respiration

(3) External Fertilization

(4) Limbless body

146. Catalase possesses 'X' as its prosthetic group. 'X' is also present in

(1) Peroxidase

(2) Carbonic anhydrase

(3) Carboxypeptidase

(4) Succinic dehydrogenase

Vivek Krishna

147. The gas exchange systems of aquatic arthropods, molluscs as well as birds were compared and it was concluded that the common feature among all the gas exchange systems is that they are

(1) Maintained at a constant temperature

(2) Moist

(3) Enclosed within ribs

(4) Exposed to air

148. Match **Column I** with **Column II** and select the correct option w.r.t. humans under normal physiological conditions.

	Column I		**Column II**
a.	Atrial systole	(i)	First heart sound, lub, is heard at its beginning
b.	Ventricular systole	(ii)	Forces blood *via* the open AV valves into the ventricles
c.	Atrial diastole	(iii)	Second heart sound, dub, is heard at its beginning
d.	Ventricular diastole	(iv)	Blood is collected in atria

(1) a(iii); b(ii); c(iv); d(i)

(2) a(i); b(ii); c(iii); d(iv)

(3) a(ii); b(i); c(iv); d(iii)

(4) a(iv); b(iii); c(i); d(ii)

149. Choose the **incorrect** match w.r.t humans.

(1)	Liver	–	Secretes bile juice containing bile salts
(2)	Lungs	–	Remove approximately 12 mL of CO_2 per hour and significant quantities of water
(3)	Sweat glands	–	Produce sweat, whose primary function is to facilitate a cooling effect on the body surface
(4)	Sebaceous glands	–	Eliminate substances such as sterols, hydrocarbons and waxes *via* sebum

(1) (1)

(2) (2)

(3) (3)

(4) (4)

150. In humans, the type of joint present between the 1st and 2nd cervical vertebrae is _____ joint while the type of joint present between the carpals is _____ joint.
Select the option that correctly fills the respective blanks.

(1) Gliding, saddle

(2) Cartilaginous, hinge

(3) Saddle, hinge

(4) Pivot, gliding

151. Choose the incorrect statement w.r.t. impulse conduction along a nerve fibre.

(1) Electrical potential difference across the resting plasma membrane is called resting membrane potential

(2) Na^+ influx makes the inside of axonal membrane negative as compared to the outside

(3) During depolarisation, polarity of resting axonal membrane is reversed

(4) The rise in the stimulus induced permeability to Na^+ is extremely short-lived.

152. All of the given functions are performed by the hormone secreted from the endocrine gland located on the dorsal side of forebrain, **except**

(1) Regulation of diurnal rhythm of the body

(2) Increasing Ca^{2+} absorption from the digested food

(3) Maintenance of body temperature

(4) Regulation of pigmentation as well as our defence capability

153. In the case of embryonic development w.r.t. humans, by the end of how many weeks, the body of foetus is covered with fine hair and eyelashes are formed?

(1) 24

(2) 12

(3) 15

(4) 10

154. Despite of having unprotected sexual intercourse for about 2 years, the female partner of the couple was unable to conceive. They went to the doctor for checkup and found that the female was unable to produce ova while the male partner was normal. Thus, the doctor suggested a method that involves *in-vivo* fertilisation. Identify that method among the following that is most likely suggested by doctor and select the correct option.

(1) ZIFT

(2) IUT

(3) AI

(4) GIFT

Vivek Krishna

155. Eye of the *Octopus* and mammals are the example of 'P' structures, which are the result of 'Q' evolution.
which of the following do not exemplify 'Q' evolution.
(a) Lemur and spotted cuscus
(b) Wombat and Bandicoot
(c) Flippers of penguins and dolphins
(d) Flying phalanger and koala
Choose the correct option.

(1) (a) and (d)

(2) (b) and (c)

(3) (b) and (d)

(4) (a) and (c)

156. The species-area relationship is described by equation $S = CA^Z$. If C = 5, Z = 0.25, area of island-1 = 16 km^2 and area of island-2 = 256 km^2, what would be the ratio of species richness of island 2 : island 1

(1) 2 : 1

(2) 4 : 1

(3) 1 : 2

(4) 1 : 4

157. During lactic acid fermentation, which enzyme participates in the conversion of pyruvic acid to lactic acid?

(1) Lactate oxidase

(2) Lactate dehydrogenase

(3) Pyruvate dehydrogenase

(4) Pyruvate decarboxylase

158. Read the following Assertion (A) and Reason (R) and select the correct option
Assertion (A): At certain concentration of ethanol it becomes poisonous to yeast.
Reason (R): The ethanol production by *Saccharomyces cerevisiae* reduces after a certain concentration of ethanol is reached.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

159. Tomato and maize are grown in a greenhouse while maintaining the temperature at 38°C.
It was observed that the maize productivity surpasses the productivity of tomato. Select the most appropriate option for the given observation.

(1) In tomato, the enzymes that participate in dark reaction are much sensitive to high temperature than that in maize

(2) Maize shows double CO_2 fixation while tomato does not.

(3) Both benefit equally from CO_2 but the optimum temperature required for tomato was not achieved

(4) Both benefit equally from temperature but CO_2 concentration became limiting to tomato.

160. Which of the following occurs in mitochondria when ATP synthesis takes place through oxidative phosphorylation?

(1) Increase in pH of intermembrane space

(2) Transport of electrons from one complex to another

(3) Reduction of NAD^+ into $NADH + H^+$

(4) Pumping of protons against concentration gradient into the intermembrane space

161. Two different plants were treated with the hormones.
Plant A: Treated with a hormone that increases the length of the stem to increase its yield.
Plant B: Treated with a hormone that promotes the female flowering, thus increasing the yield.
Identify the **correct** statement for the given observation?

(1) Plant A can be sugarcane which is treated with the gaseous hormone.

(2) Plant B can be cucumber which is treated with the ethephon

(3) Plant A can be sugarcane which is treated with carotenoid derived hormone.

(4) Plant B can be cucumber which is treated with ethylene

162. Which of the following statements about the Meselson-Stahl's experiment is **incorrect**?

(1) It proved semiconservative mode of replication of DNA

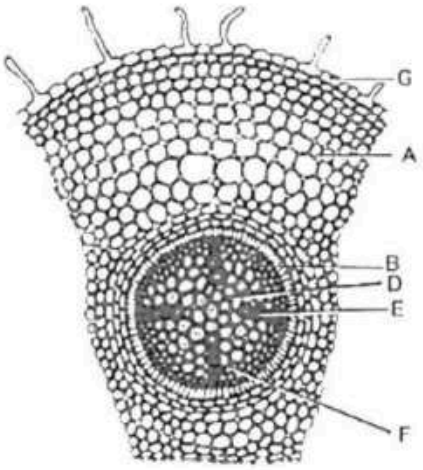
(2) Heavy nitrogen (^{15}N) was used to label parental DNA

(3) After two generations in ^{14}N medium, only light DNA was observed

(4) The DNA band obtained after one generation was of intermediate density

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163.



Consider the above diagram and choose the **correct** option

(1) Tissues labelled as D, E and F constitute the stele

(2) Tissues labelled as A, E, F and G perform the function of photosynthesis

(3) Tissues labelled as A, B and G are made up of dead cells

(4) Tissues labelled as A, F and G can perform the role of photosynthesis

164. Slime moulds, has a life cycle involving spore-like stages. Which of the following explains the advantage of this stage?

(1) Enables free-living and creeping

(2) Survive adverse environmental condition and aids transmission

(3) Facilitates feeding via pseudopodia

(4) Allows reproduction by binary fission in water

165. In a logistic growth curve, the population stabilizes at carrying capacity (K) because

(1) Birth rate > Death rate

(2) Birth rate < Death rate

(3) Birth rate = Death rate

(4) Immigration > Emigration

166. Which among the following combinations is **not** correctly matched.

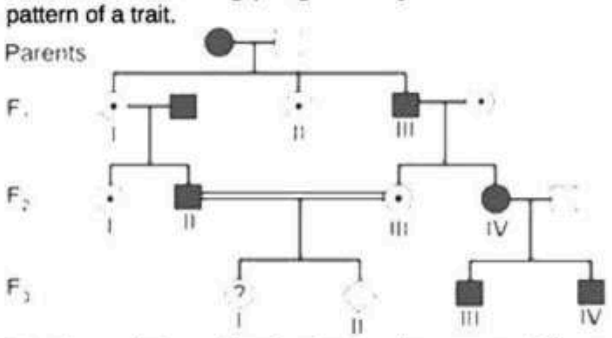
(1) Systematics – Takes into account evolutionary relationship between organisms.

(2) Classification – Grouping based only on morphological characters

(3) Taxonomy – Identification, nomenclature and classification

(4) Nomenclature – Assigning scientific name

167. Consider the following pedigree analysis for an inheritance pattern of a trait.



Certain symbols used in the given pedigree are as follows.

– Normal male

– Normal female

– Affected male

– Affected female

– Sex unspecified

– Carrier female

(A) All the female progeny in the F_1 generation are carrier for the trait.

(B) If individual I of F_2 generation is married to a heterozygous individual for sickle cell anemia then probability of their sons to be affected is 50%.

(C) The given pedigree is true for family exhibiting myotonic dystrophy, haemophilia or thalassemia.

(D) Consanguineous mating in the pedigree resulted in the birth of individual afflicted with X-linked disorder.

(E) The probability of the individual II in F_3 generation of being affected male is 25%

Segregate the statements in the column of True and False and choose the correct option.

	True	**False**
(1)	B and C	A, D and E
(2)	A and E	B, C and D
(3)	A, B and E	C and D
(4)	A, D and E	B and C

(1) (1)

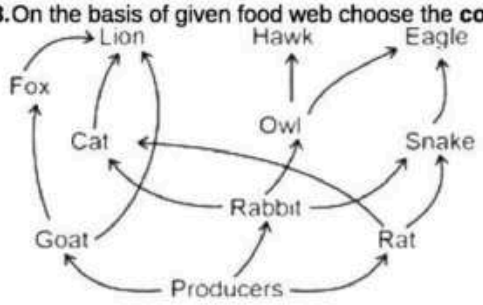
(2) (2)

(3) (3)

(4) (4)

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168. On the basis of given food web choose the **correct** option.



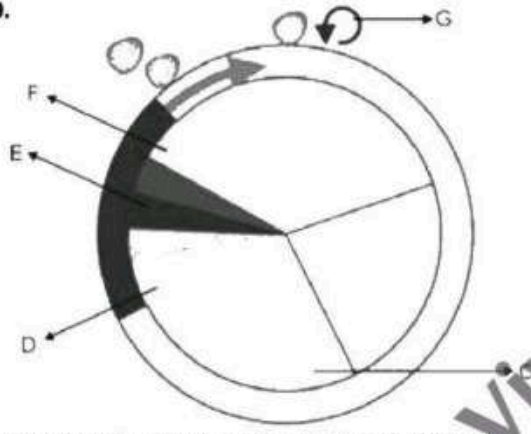
(1) The energy at the level of plants will be 10 times more as compared to the level of hawk

(2) The amount of energy loss from T_2 to T_3 level is 0.9 kJ if 10000 kJ energy in the form of solar radiation falls on the producers.

(3) The given food web is composed of more than three food chains and none of them will form spindle shaped pyramid of number

(4) In the given ecosystem, none of the mentioned organisms is occuping more than one trophic level.

169.



Consider the above representation of cell cycle.
(i) Events taking place during phase 'D' in mitosis is similar to the events taking place in first phase of meiosis I.
(ii) Only the double membrane bound cell organelles get duplicated during phase 'C'.
(iii) The cells exit from a particular phase of cell cycle to enter an metabolically inactive stage is labelled as G
(iv) During phase 'E', the centromere of each homologous chromosome get directed towards the same pole and the arms trail behind it in this sub-stage of meiosis
(v) In an organism without cell wall, the event labelled as F can be achieved by the appearance of a furrow in plasma membrane.
How many of the above statements are **incorrect**?

(1) Three

(2) Four

(3) Two

(4) One

170. Read the following statements carefully and identify the one that is **not** true.

(1) Materials to be packaged in the form of vesicles from the endoplasmic reticulum fuse with the concave face of the Golgi apparatus and then move towards the convex face.

(2) The endoplasmic reticulum which are extensive and continuous with the outer membrane of the nucleus, is frequently observed in the cells that are actively involved in protein synthesis.

(3) In an endomembrane system, Golgi apparatus remains in close association with endoplasmic reticulum.

(4) Proteins synthesised by ribosomes on the ER are modified in the cisternae of Golgi apparatus.

171. How many of the diseases given in the box below is/are considered as viral air-borne disease(s)?

Common cold, Typhoid, Plague, Dengue, Diptheria

Select the correct option.

(1) Three

(2) Two

(3) One

(4) Four

172. Choose the **correct** match.

(1)	Recombinant insulin	–	Possesses A, B and C peptide chains
(2)	Adenosine deaminase	–	Crucial for immune system to function
(3)	Recombinant therapeutics	–	30 of these are being marketed in India
(4)	GM plants	–	Reduced efficiency of mineral usage

(1) (1)

(2) (2)

(3) (3)

(4) (4)

173. Select the **correct** statement.

(1) Bacterial cells are made competent by treating them with a specific concentration of a monovalent anion.

(2) To introduce a particular piece of DNA into an animal cell, use of *Agrobacterium* is the most suitable method.

(3) Restriction endonuclease cuts each of the two strands of the DNA double helix at specific points in their sugar-phosphate backbones.

(4) DNA that precipitates after addition of chilled ethanol is removed by elution.

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174. Parturition is induced by a complex neuroendocrine mechanism that involves all of the given hormones, **except**

(1) Hormone which is the main glucocorticoid in our body

(2) Hormone which is also known as the milk-ejecting hormone

(3) Hormone which is predominantly released by the growing ovarian follicles

(4) Hormone which is released by pars distalis and regulates growth of mammary glands.

175. Match Column I with Column II and select the correct option.

	Column I		**Column II**
a.	*Naja*	(i)	Homeotherm and exhibits heterodont dentition
b.	*Columba*	(ii)	Possesses air bladder and operculum
c.	*Canis*	(iii)	External ear openings are absent
d.	*Clarias*	(iv)	Has additional chambers in its digestive tract

(1) a(iii); b(iv); c(i); d(ii)

(2) a(i); b(ii); c(iii); d(iv)

(3) a(iv); b(iii); c(ii); d(i)

(4) a(ii); b(iv); c(iii); d(i)

176. Read the given statements and select the **correct** option.
Statement (A) : Stirred-tank bioreactors have been designed for ensuring anaerobic conditions in the culture vessel.
Statement (B) : The downstream processing and quality control testing vary from product to product.

(1) Both the statements (A) and (B) are correct

(2) Both the statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

177. How many of the given statements is/are correct?
(a) All the homopolysaccharides whose monomeric unit is glucose are considered as secondary metabolites.
(b) Adult human haemoglobin exhibits quaternary structure and consists of 4 subunits.
(c) The effect of competitive inhibitor cannot be reversed by adding more substrate.
Choose the **correct** option.

(1) Two

(2) One

(3) Three

(4) Zero

178. Select the **correct** match w.r.t evolution of animals.

(1) Thecodonts – Ancestors of snakes and tuataras

(2) Therapsids – Predecessors of sauropsids

(3) Pelycosaurs – Descendants of synapsids

(4) Synapsids –Gave rise to crocodiles and birds

179. In a human, under normal physiological conditions, if the blood entering in descending limb of vasa recta has the osmolarity of 300 mOsmol/L, then, what will be the approximate osmolarity of the blood just leaving the ascending limb of vasa recta at the junction of cortex and medulla?

(1) 1200 mOsmol/L

(2) 800 mOsmol/L

(3) 300 mOsmol/L

(4) 600 mOsmol/L

180. In a hypothetical condition of a person, if total ATP molecules are removed from the body, then what would be the effect on a contracted muscle?

(1) Muscle would relax immediately

(2) Muscle would still contract and relax at the normal rate

(3) Muscle would remain in a rigid and contracted state

(4) Muscle would first relax and then contract abruptly at a faster rate.

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